

Afshin Malek

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TECHNICAL SUMMARY

Computer Science graduate student (4.00 GPA across post-baccalaureate STEM and upper-division CS coursework) specializing in C++ and Java systems programming, data structures and algorithms, multithreaded concurrency, socket networking, and full-stack web architecture.

TECHNICAL SKILLS

Languages: C++, C, Java, Python, JavaScript (ES6+), x86 / MIPS Assembly, SQL, Bash, HTML5, CSS3

Systems & Architecture: Object-Oriented Design, Data Structures & Algorithms, Computer Architecture, Memory Management, Linux/POSIX CLI, File I/O Streams, Concurrency, Networking

Web & Frameworks: React, Redux Toolkit, React Router, Node.js, Express.js, MongoDB, Mongoose, RESTful APIs, JSON, Axios, Nodemailer, Passport

Developer Tools: Git, GitHub, Make/CMake, VS Code, GDB

Mathematics: Discrete Mathematics, Linear Algebra, Multivariable Calculus

EDUCATION

California State University, East Bay Aug 2026 - Present

Graduate Computer Science Bridge (Conditional M.S.)

GPA 4.00/4.00. CS 301 Data Structures & Algorithms (A), CS 321 Computer Organization & Architecture (A).

Spring 2027: CS 421 Operating Systems, CS 413 Analysis of Algorithms.

Peralta Community College District & De Anza College Aug 2024 - May 2026

Post-Baccalaureate CS & Mathematics

GPA 4.00/4.00, 45 Peralta units, Dean's List, Certified Peer Tutor; x86 Assembly at De Anza College, A+.

San Francisco State University Jan 2014 - Jan 2016

B.S. Business Administration (Finance)

GPA 3.34, Dean's List (2x).

WORK EXPERIENCE

Certified Computer Science & STEM Peer Tutor Jul 2025 - May 2026

Berkeley City College, Berkeley, CA

• Guided undergraduate students through C++ and Java object-oriented design, pointer arithmetic, manual memory management, recursion, data structures, asymptotic analysis, Multivariable Calculus, and Linear Algebra.

• Conducted one-on-one code reviews and live debugging sessions focusing on compiler diagnostics, GDB troubleshooting, memory-leak prevention, and modular code architecture.

Full-Stack Web Developer Jan 2023 - Jun 2025

Self-Employed (Freelance), San Francisco Bay Area, CA

• Architected and deployed full-stack MERN applications (MongoDB, Express.js, React, Node.js) featuring responsive single-page UIs, Redux Toolkit state management, and client-side routing.

• Engineered RESTful APIs with Google OAuth/Passport authentication, Mongoose schema validation, and compound database indexing to enforce data integrity.

TECHNICAL PROJECTS

File Processing Utilities C++, File I/O, Stream Parsing

github.com/afshinmalek/file-processing

20 single-pass streaming utilities built on ifstream and stringstream, parsing CSV, multi-delimiter, key=value configuration, fixed-width, and quoted-field records without whole-file loading. Header-aware column aggregation, predicate-based filtering, and extremum searches, with runtime_error handling for missing files and columns.

C++ Spaced Repetition Study Engine C++, OOP, File I/O, CLI

github.com/afshinmalek/spaced-repetition-quiz-engine

Streak-based mastery scheduling: low-streak questions repeat, a streak of 2+ means mastered; encapsulated class hierarchies, file-stream persistence, and strict input validation.

Battleships Java, Swing, Networking

github.com/afshinmalek/battleships

Socket-based client/server Battleships with 25x25 player and tracking grids, deterministic ship placement, and synchronized turn state. Hit/miss/sunk/win protocol for shot resolution and grid updates between client and server.

Fork/Join Distance Calculator Java, Concurrency

github.com/afshinmalek/fork-join-distance-calculator

Fork/Join parallelism benchmarked against single-threaded execution on 10 million point pairs. Performance compared at several split thresholds.

Supplier-Consumer Threads Java, Concurrency

github.com/afshinmalek/supplier-consumer-threads

Bounded-buffer thread synchronization: ten suppliers and ten consumers coordinated via a shared monitor using synchronized blocks and wait/notify signaling to prevent deadlock and race conditions, with timestamped logging.

Red-Black BST Java, Data Structures

github.com/afshinmalek/red-black-bst

Generic self-balancing red-black binary search tree with guaranteed O(log N) insertion, deletion, and in-order traversal, using left/right rotations and color-flip rebalancing.

Dijkstra Shortest Path Java, Graphs, Priority Queues

github.com/afshinmalek/dijkstra-shortest-path

Dijkstra's shortest-path algorithm over an indexed min-priority queue: O((V+E) log V) shortest-path routing with multi-source facility assignment on a 50-node weighted graph.

Lazy Prim's MST - Minimum Cable Network Java, Graphs

github.com/afshinmalek/lazy-prim-mst

Lazy Prim's minimum spanning tree for cable-network optimization, with per-station MST costs to choose the cheapest field station.

BookMe - MERN Full-Stack Booking App React, Redux Toolkit, Express.js, Node.js, MongoDB

github.com/afshinmalek/bookedCRA-client | github.com/afshinmalek/bookedCRA-server

Component-driven React single-page application with Redux Toolkit state management, React Calendar booking, and client-side routing; modular Express/Node.js services exposing CRUD RESTful endpoints. Google OAuth via Passport; MongoDB with Mongoose modeling and a unique date + time index that blocks double-bookings; Nodemailer confirmation emails.

Game of Life Java, Swing

github.com/afshinmalek/game-of-life

Swing GUI for Conway's Game of Life with a threaded generation loop. User-chosen grid size for the simulation.